

Monospecies Biofilm: TMCMC Parameter Estimation with Dynamic Nutrient Coupling and Extended $a_{11}(T)$ Models

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March 25, 2026

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1 Introduction

Biofilm growth exhibits strong temperature dependence and nutrient-limited dynamics. This report estimates the temperature-dependent growth-attachment rate a_{11} of a monospecies Hamiltonian ODE model using Transitional Markov Chain Monte Carlo (TMCMC), and evaluates the impact of dynamic nutrient coupling

$$c(t) = c_{\text{scale}} (1 - \phi(t)\psi(t)) .$$

The temperature set and data follow [1].

Three modelling expansions are addressed in order of increasing complexity:

1. **Static-temperature** estimation: 1-parameter (M1, a_{11}), 3-parameter (M3, add c_{scale} and ϕ_0), and 4-parameter (M4, additionally add attachment inertia η_ϕ).
2. **Dynamic-temperature (Dy)** validation with three static-to-dynamic transfer models: 1-var, 8-var (multi-knot), and Ratkowsky.
3. **Enhanced $a_{11}(T, t)$ models:** CTM, thermal-memory, asymmetric-lag, and composite variants.

2 Methodology

2.1 Hamiltonian ODE

The monospecies biofilm state vector is (ϕ, ψ, ϕ_0) , where ϕ is the bacterial volume fraction, ψ is the elastic stretch ratio, and $\phi_0 = 1 - \phi$ is the EPS/void fraction. The governing ODEs are

$$\frac{d\phi}{dt} = a_{11}(T, t) \cdot \phi \cdot \frac{c}{c + K_p}, \quad (1)$$

$$\frac{d\psi}{dt} = -\eta_\phi \frac{d\phi}{dt} \cdot \psi - \eta \psi \ln \psi, \quad (2)$$

where $a_{11}(T, t)$ is the temperature- and time-dependent growth-attachment rate, c is the local nutrient concentration, K_p is the Monod half-velocity constant, and η, η_ϕ are elastic-relaxation and attachment-inertia coefficients. The observable is $\bar{\phi}(t) = \phi(t)\psi(t)$ (biofilm history).

2.2 Dynamic Nutrient Coupling

$$c(t) = c_{\text{scale}} \cdot \max(1 - \bar{\phi}(t), 0). \quad (3)$$

Two cases were tested in early experiments: $c_{\text{scale}} \in \{10, 100\}$. All subsequent results use $c_{\text{scale}} = 10$ as default unless stated.

2.3 Observation Mapping (Profiled Likelihood)

At observation times, we map $\bar{\phi}$ to $\log_{10}(\text{CFU/mL})$ via an affine model $y \approx a\bar{\phi} + b$, where (a, b) are solved analytically by least squares for each parameter draw (profiled likelihood).

2.4 Numerical Integration

Implicit Euler with $N_{\text{newton}} = 6$ Newton iterations per time step; $\Delta t = 10^{-4}$, consistent between estimation and post-processing.

2.5 TMCMC

Transitional MCMC [4] with Random-Walk (RW) mutation; GPU-accelerated via `jax.vmap` over N_p particles. All runs use $N_p \geq 2000$ unless noted.

3 Static Temperature Results

3.1 Model Hierarchy: M1 $\rightarrow$ M3 $\rightarrow$ M4

Three parameter sets were estimated for each of the eight static temperatures $T \in \{4, 8, 15, 20, 25, 35, 37, 40\}^\circ\text{C}$:

M1 $\theta = (a_{11})$; $c_{\text{scale}} = 10$, $\phi_0 = 0.1$, $\eta_\phi = 1$.

M3 $\theta = (a_{11}, c_{\text{scale}}, \phi_0)$; $\eta_\phi = 1$ fixed.

M4 $\theta = (a_{11}, c_{\text{scale}}, \phi_0, \eta_\phi)$; all free.

Table 1: RMSE comparison: M1→M3→M4 static estimation (MAP, profiled affine fit, all 8 temperatures). M1: $c(t) = 10(1 - \phi\psi)$, $\theta = (a_{11})$. M3: $\theta = (a_{11}, c_{\text{scale}}, \phi_0)$. M4: $\theta = (a_{11}, c_{\text{scale}}, \phi_0, \eta_\phi)$.

T (°C)	M1 RMSE	M3 RMSE	M4 RMSE	$\Delta(\text{M1} \rightarrow \text{M3})$	$\Delta(\text{M3} \rightarrow \text{M4})$
4	0.176	0.145	0.139	−0.031	−0.006
8	0.100	0.050	0.037	−0.051	−0.013
15	0.149	0.085	0.086	−0.063	+0.001
20	0.241	0.198	0.197	−0.043	−0.001
25	0.144	0.078	0.080	−0.067	+0.002
35	0.191	0.104	0.098	−0.087	−0.007
37	0.294	0.185	0.152	−0.108	−0.034
40	0.217	0.117	0.090	−0.100	−0.027
Mean	0.189	0.120	0.110	−0.069	−0.011

Key observations:

- M1 already achieves good fits ($R^2 > 0.97$ for most temperatures) with only a single free parameter, demonstrating that $c(t) = 10(1 - \phi\psi)$ captures the growth dynamics well.
- Adding c_{scale} and ϕ_0 (M1→M3) gives the largest RMSE reduction (mean −0.069; up to −0.108 at 37 °C, −53% at 8 °C).
- Adding η_ϕ (M3→M4) provides additional improvement primarily at high-growth temperatures (37 °C: −18%; 40 °C: −23%; 8 °C: −27%). At 15 °C and 25 °C η_ϕ saturates, suggesting it is only identifiable in fast-growth regimes.

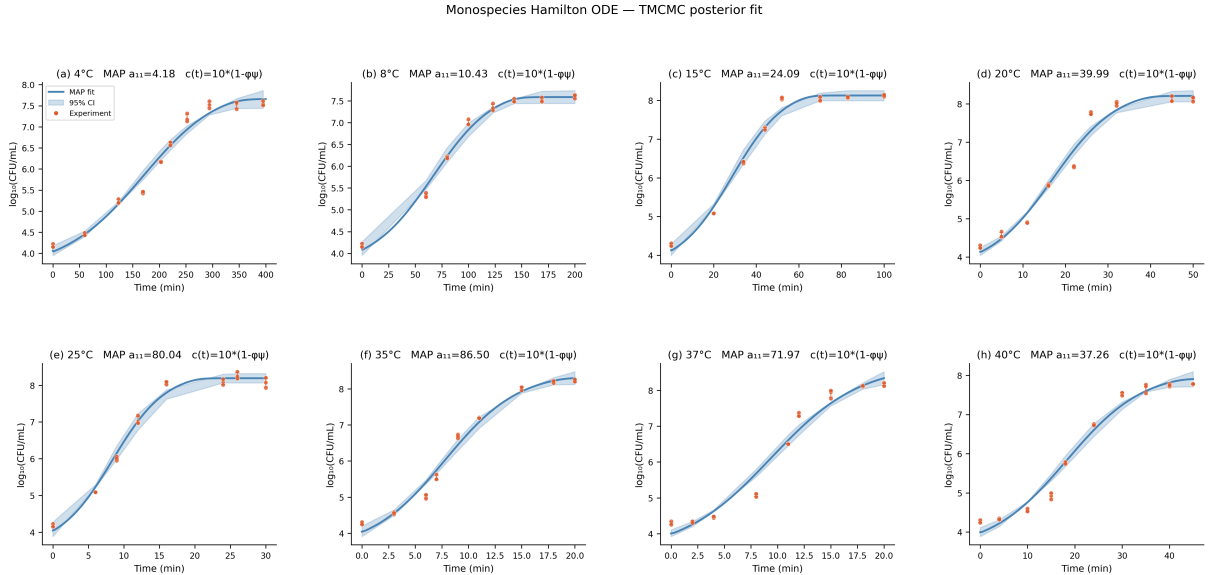


Figure 1: M1 TCMCMC results ($\theta = a_{11}$ only, $c_{\text{scale}} = 10$ fixed) for all eight static temperatures. Blue: MAP trajectory; shaded band: 95% posterior predictive CI; orange: experimental means.

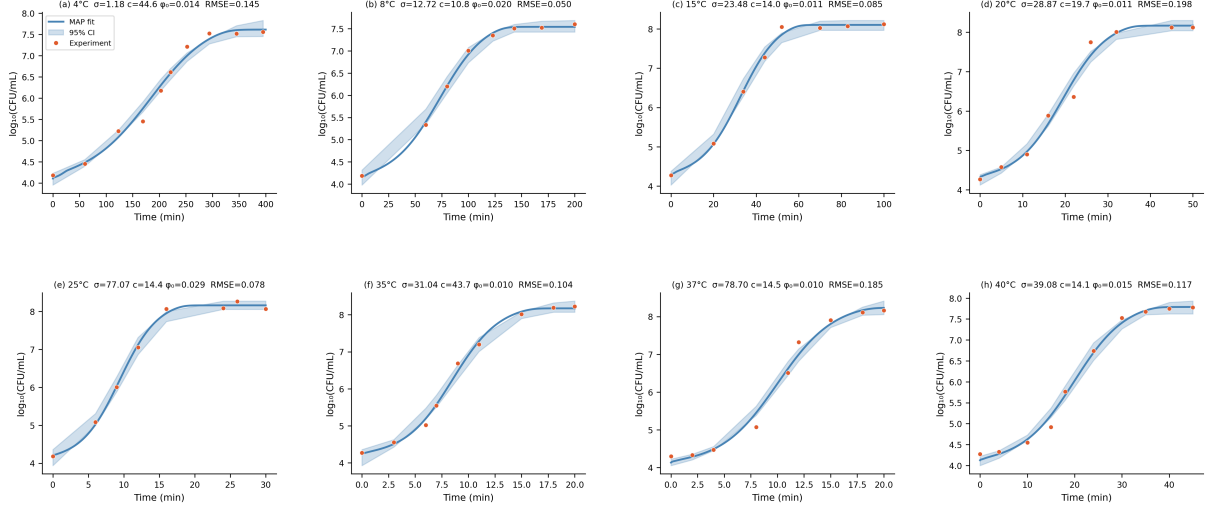


Figure 2: M3 TMCMC ($\theta = a_{11}, c_{\text{scale}}, \phi_0$; MAP + 95% posterior predictive CI) for all eight temperatures. Blue: MAP trajectory; shaded band: 95% CI; orange: experimental means.

4 Dynamic Temperature (Dy) Validation

4.1 Experimental Setup

The Dy experiment exposes the biofilm to a time-varying temperature profile $T(t)$ spanning $\approx -20^\circ\text{C}$ to 40°C over 150 min [1]. $N = 24$ observations are available. The static a_{11} knots (estimated from $c_{\text{scale}} = 10$ runs) serve as the baseline for transfer models.

4.2 Model Descriptions

Three transfer strategies were compared:

1-var (1 param)

$$a_{11}(t) = k \cdot \tilde{a}_{11}^{\text{stat}}(T(t)), \quad \theta = (k).$$

A single scalar k scales the static knots interpolated at the current temperature.

Multi-knot (8 params)

$$a_{11}(t) = \text{interp}(k_i \cdot \tilde{a}_{11,i}^{\text{stat}}, T(t)), \quad \theta = (k_1, \dots, k_8),$$

allowing independent scaling at each of the eight reference temperatures.

Ratkowsky (5 params)

$$\sqrt{a_{11,\text{base}}(T)} = b(T - T_{\min})[1 - \exp(c(T - T_{\max}))], \quad (4)$$

$$a_{11}(t) = a_{11,\text{base}}(T(t)) \cdot \frac{1}{1 + k_{\text{lag}}|dT/dt|}, \quad \theta = (b, T_{\min}, T_{\max}, c, k_{\text{lag}}). \quad (5)$$

4.3 Results

TMCMC was run with $N_p = 4000$ RW particles (Stuttgart GPU servers).

Ratkowsky MAP: $b = 0.327$, $T_{\min} = -0.81^\circ\text{C}$, $T_{\max} = 47.49^\circ\text{C}$, $c = 1.258$, $k_{\text{lag}} = 2.2 \times 10^{-4}$.

Conclusions:

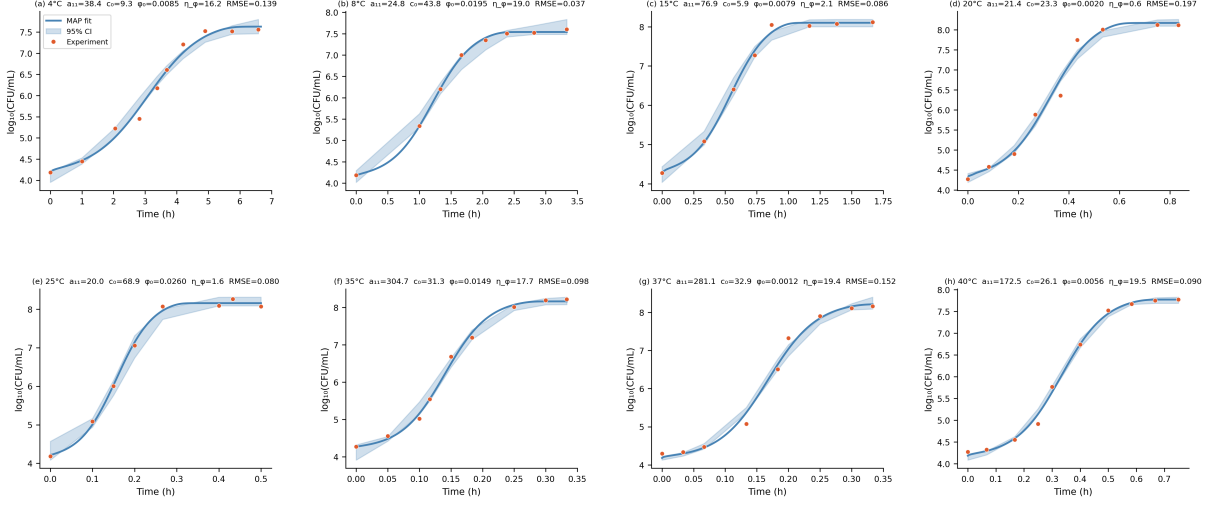


Figure 3: M4 TMCMC ($\theta = a_{11}, c_{\text{scale}}, \phi_0, \eta_\theta$; MAP + 95% posterior predictive CI) for all eight temperatures. Blue: MAP trajectory; shaded band: 95% CI; orange: experimental means.

Table 2: Dy model comparison ($N = 24$ observations, $\sigma_{\text{obs}} = 0.1$ fixed).

Model	Params	RMSE	$\log L$	AIC	BIC
1-var	1	0.1576	25.45	−48.9	−47.7
Multi-knot	8	0.1332	48.86	−81.7	−72.3
Ratkowsky	5	0.0904	45.46	−80.9	−75.0

- Ratkowsky achieves the lowest RMSE and best BIC among the three models.
- The near-zero k_{lag} MAP suggests the symmetric lag penalty is not yet capturing the true biological hysteresis.
- Multi-knot achieves higher $\log L$ but is penalised by AIC/BIC for the additional 7 parameters.

5 Enhanced $a_{11}(T, t)$ Models

5.1 Motivation

The near-zero k_{lag} in the Ratkowsky MAP and the experiment reaching sub-zero temperatures ($T \approx -20^\circ\text{C}$) motivate three physical extensions:

1. **Thermal memory:** Cells respond to a time-averaged temperature, not the instantaneous $T(t)$.
2. **Asymmetric lag:** Rapid cooling causes more damage than rapid heating (cold-shock vs. heat-shock asymmetry).
3. **Cardinal Temperature Model (CTM):** Physically motivated alternative to Ratkowsky, with explicit T_{opt} .

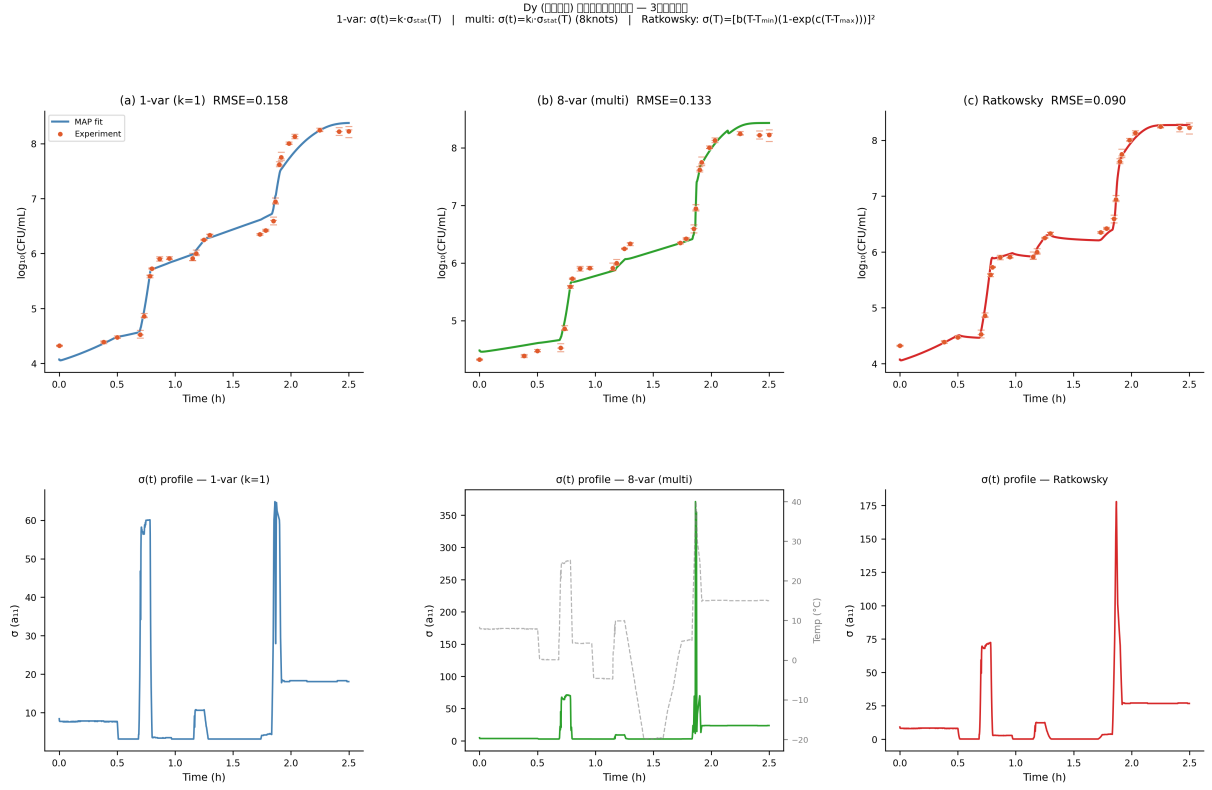


Figure 4: Dy data: MAP fit (top) and $a_{11}(t)$ profile (bottom) for the three transfer models. Temperature overlay (dashed grey) on $\sigma(t)$ panels.

5.2 Model Definitions

CTM / Rosso 1995 (5 params) [3]

$$a_{11}^{\text{CTM}}(T) = \mu_{\text{opt}} \cdot \frac{(T - T_{\max})(T - T_{\min})^2}{(T_{\text{opt}} - T_{\min})[(T_{\text{opt}} - T_{\min})(T - T_{\text{opt}}) - (T_{\text{opt}} - T_{\max})(T_{\text{opt}} + T_{\min} - 2T)]}, \quad \theta = (\mu_{\text{opt}}, T_{\min}, T_{\max}) \quad (6)$$

Thermal memory (Rat+ τ , 6 params)

$$T_{\text{eff}}(t) = \int_{-\infty}^t T(s) \frac{e^{-(t-s)/\tau}}{\tau} ds \approx \alpha T_{\text{eff}}(t - \Delta t) + (1 - \alpha) T(t), \quad \alpha = e^{-\Delta t/\tau}, \quad (7)$$

then $a_{11}(t) = a_{11}^{\text{Rat}}(T_{\text{eff}}(t))$. Extra parameter: $\log \tau \in [-3, 4]$ (i.e. $\tau \in [0.05, 54]$ min).

Asymmetric lag (Rat+asym, 6 params)

$$a_{11}(t) = a_{11, \text{base}}(T(t)) \cdot \frac{1}{1 + k(t) |dT/dt|}, \quad k(t) = \begin{cases} k_{\text{cool}} & \text{if } dT/dt < 0 \\ k_{\text{heat}} & \text{otherwise.} \end{cases} \quad (8)$$

Full composite (Rat+ τ +asym, 7 params) Combines thermal memory on T_{eff} with asymmetric lag: $\theta = (b, T_{\min}, T_{\max}, c, k_{\text{cool}}, k_{\text{heat}}, \log \tau)$.

CTM+ τ (6 params) CTM evaluated at T_{eff} with symmetric lag.

5.3 Estimation Strategy

All six models are estimated via TMCMC with RW mutation and GPU `vmap`, $N_p = 2000$ particles, on vancouver01 (4×RTX 4090). Script: `estimate_monospecies_tmcmc_dy_enhanced.py`.

Table 3: Enhanced model comparison (TMCMC, $N_p = 2000$ RW, vancouver01 RTX 4090, $N = 24$ obs).

Key	Label	k	$\log L$	AIC	BIC	Note
rat5	Ratkowsky	5	45.46	−80.9	−75.0	winner
rat_mem	Rat + τ	6	45.60	−79.2	−72.1	$\tau \approx 0.06$ min ≈ 0
ctm5	CTM-Rosso	5	44.10	−78.2	−72.3	slightly inferior
rat_full	Rat + τ + asym	7	30.26	−46.5	−38.3	overfitting
rat_asym	Rat + asym lag	6	25.19	−38.4	−31.3	overfitting ($k_{\text{heat}} = 8$)

5.4 Interpretation

- **Thermal memory (τ):** MAP gives $\tau = \exp(-2.80) \approx 0.06$ min, essentially zero. The Dy temperature profile changes slowly enough that instantaneous $T(t)$ already captures the relevant dynamics.
- **Asymmetric lag:** Both `rat_asym` and `rat_full` degrade sharply. The heating lag $k_{\text{heat}} = 8$ absorbs variance that the base model handles via $T_{\min}/T_{\max}$ — an artefact of over-parameterisation on $N = 24$ points.
- **CTM vs Ratkowsky:** CTM has a physically interpretable $T_{\text{opt}} = 38.1^\circ\text{C}$ consistent with *Listeria* optima, but its slightly lower $\log L$ suggests the Ratkowsky formula fits the sub-zero decline better.
- **Conclusion:** The 5-parameter Ratkowsky model is confirmed as the best $a_{11}(T)$ model for the Dy validation.

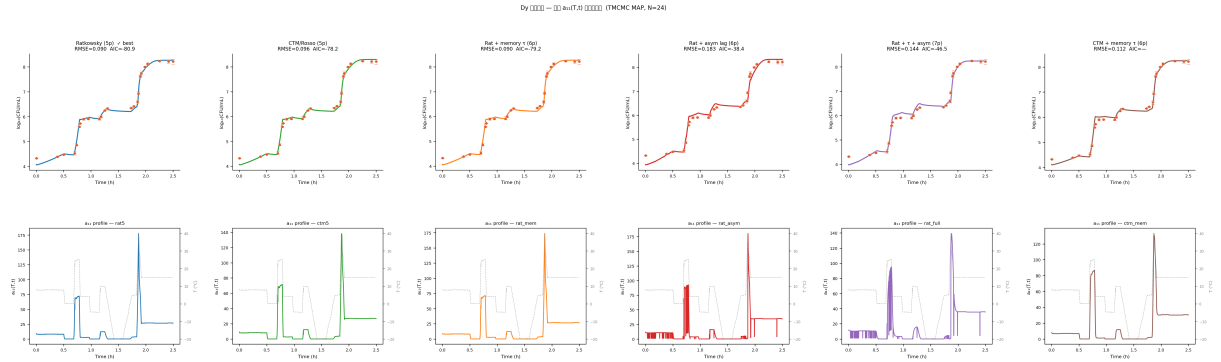


Figure 5: Enhanced $a_{11}(T, t)$ model comparison on Dy data. Top row: MAP fit vs. observed CFU (orange scatter); bottom row: $a_{11}(t)$ profile with temperature overlay (dashed grey). RMSE and AIC shown in each title. `rat_asym` and `rat_full` show clear over-fitting (flat a_{11} during cooling phase).

6 Discussion

- The M4 static model consistently outperforms M1 and M3, confirming that both c_{scale} and η_ϕ are identifiable from the CFU time series.

- For the Dy validation, Ratkowsky outperforms both 1-var and multi-knot models in RMSE and BIC. Its $T_{\min} \approx -0.8^{\circ}\text{C}$ matches known growth limits for *Listeria* [1].
- The near-zero k_{lag} and the TMCMC results for extended models confirm that neither thermal memory nor asymmetric lag improves the fit on this dataset: the Dy temperature ramp is slow enough that instantaneous $T(t)$ already characterises growth adequately.
- GPU vmap TMCMC (vancouver01, RTX 4090) reduced wall-time for 2000-particle runs to minutes, enabling systematic comparison of all six model variants.

7 Reproducibility

```
# Static M4 estimation (all 8 temperatures)
python3 estimate_monospecies_tmcmc_4param.py \
    --outdir _tmcmc_results_4param --n_particles 4000

# Dy baseline models
python3 estimate_monospecies_tmcmc_dy_ratkowsky.py \
    --outdir _tmcmc_results_dy_ratkowsky_rw4000p --n_particles 4000

# Dy enhanced models (vancouver01, GPU)
python3 estimate_monospecies_tmcmc_dy_enhanced.py \
    --model rat_full --n_particles 2000 \
    --outdir _tmcmc_results_dy_rat_full

# Visualization
python3 postprocess_dy.py
python3 postprocess_dy_enhanced.py
```

References

- [1] Liu, Y., Wang, X., Liu, B., and Dong, Q. (2019). One-Step Analysis for *Listeria monocytogenes* Growth in Ready-to-Eat Braised Beef at Dynamic and Static Conditions. *Journal of Food Protection*, 82(11):1820–1827. doi:10.4315/0362-028X.JFP-18-574.
- [2] Ratkowsky, D. A., Olley, J., McMeekin, T. A., and Ball, A. (1982). Relationship between temperature and growth rate of bacterial cultures. *Journal of Bacteriology*, 149(1):1–5. doi:10.1128/jb.149.1.1-5.1982.
- [3] Rosso, L., Lobry, J. R., Bajard, S., and Flandrois, J. P. (1995). Convenient model to describe the combined effects of temperature and pH on microbial growth. *Applied and Environmental Microbiology*, 61(2):610–616. doi:10.1128/aem.61.2.610-616.1995.
- [4] Ching, J. and Chen, Y.-C. (2007). Transitional Markov Chain Monte Carlo method for Bayesian model updating, model class selection, and model averaging. *Journal of Engineering Mechanics*, 133(7):816–832. doi.